European Developer Conference



New Technologies, New Opportunities

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Developer Architecture
and Disclosure



What's new?

Hardware: 68k, many ARM chips

Screens: sizes, AIA, Double density, 1.5x density

Networking: Bluetooth, 802.11, GSM, GPRS…

Market focuses: wireless voice and/or data, gaming, location-based, ultra-mobile, education…

New formfactors: bigger, smaller, "different"…

Much more memory, inexpensive VFS…

Localizations: languages, encodings…

Input methods: keys, microphones, Graffiti® 2…

The Diversity of Palm OS® devices



Diversity



Diversity can be a strength.

Diversity can be a distraction.

What's your strategy?

a) Limit your app to running on a few devices
b) Standard functionality on every device
c) Run everywhere, plus some optimization
d) 100% perfection, everywhere

Diversity in Real Life

Primary areas where diversity impacts you:

- Screens: size, density, color depth
  - (bitmaps normally should be present for 1 bit, 8 bit, and 8x2, but some applications deserve higher or lower expectations.)
- Physical buttons and keys
- Pen Input
- Memory availability, VFS cards
- Communications: Beaming, Bluetooth, IP connectivity…
- Processors: Various 68k processors, ARM, PACE
  - (usually only a problem if you're circumventing the APIs)

Diversity of Screens

Several Palm OS devices have non-square screens

- Sometimes this means a collapsible input area
- Sometimes it means landscape is possible (or normal)
- By default, applications are assumed to be ignorant, and therefore the input area will always be shown.
- This is different from screen density

Use the unified platform API and sample code

- See the kb.palmsource.com recipe (ID 507)
- Will work on all current devices, plus Palm OS 6

Diversity of Physical Buttons

Sony

- Jog dial SDK on Sony website

Palm, Inc.

- 5 way nav SDK in Dev Seeding Area

Garmin

- Header in Garmin SDK

General

- New Palm OS SDK defined generic navigation in chars.h and KeyMgr.h

Diversity of Pen Input

Graffiti

- Last shipped in Palm OS 5.1

Graffiti 2

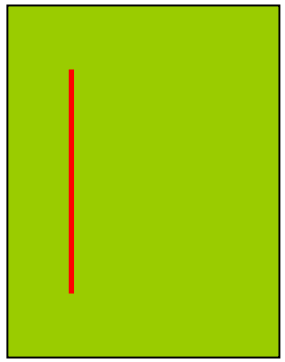
- In 5.2, two-stroke letters were complex
- As of Palm OS 5.3 it is simpler again

Other input systems

- Particularly popular in Asia

Graffiti 2

Graffiti 2 Gotchas





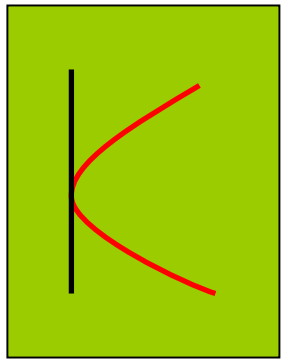
See article 397 in KB

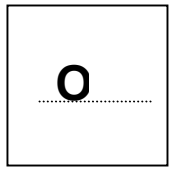
1) Sends 'L'

- **Can not use L and K as menu shortcuts**
- **Apps which process field text letter by letter need to be careful**

Graffiti 2

Graffiti 2 Gotchas





See article 397 in KB

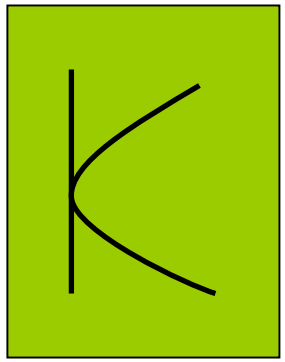
1) Sends 'L'

2) Sends backspace char

- **Can not use L and K as menu shortcuts**
- **Apps which process field text letter by letter need to be careful**

Graffiti 2

Graffiti 2 Gotchas





See article 397 in KB

1) Sends ‘L’
2) Sends backspace char
3) Sends ‘K’

- **Can not use L and K as menu shortcuts**
- **Apps which process field text letter by letter need to be careful**

Nontechnical Diversity Areas

It is not just about code:

- Performance differences (sometimes > 300x)
- Testing, support, documentation, packaging…
- Device-specific features, and varying customer profiles

Suggestions for Success

Handle the base case

- Require a reasonably current Palm OS version, etc.
- Add restrictions if necessary
  - Memory requirements are obvious
  - Device-specific features and/or hardware?

Think long-term

- Is incompatibility worth it? Perhaps…

Optimize for your target customers

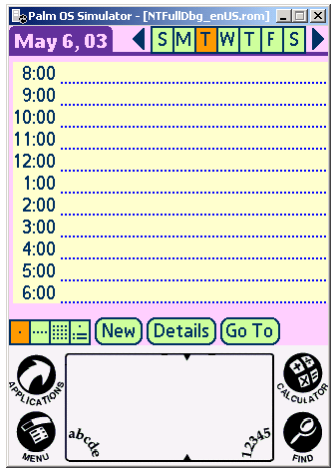
- Early adopters? Technical users? Online?
- Low cost devices? High end?
- Licensee tie-in? Controlled market?

Testing Diversity

- **Testing on actual devices**
- **Testing on emulators and simulators**
- **Testing on various networks**







Today's Opportunity

Licensees want you to optimize for their devices

- Their talks should give you guidelines
- Visit them for specifics, tools, samples…
- Try their devices in the lab
  - (Small effort here might have a big payoff)
- Work together with them on optimization and market direction

Current Affairs and Trends

Imitating desktops is easier to do

- Because of hardware improvements

Imitating desktops is harder to do *profitably*

- Simple ports of desktop applications have more competition, but less satisfaction

Current Affairs (cont.)

Devices constantly being released

- Less desire for “one size fits all” solutions
- Optimizing the user interface is an interesting coding challenge

New and easier ways to connect

- Many technologies are looking for problems to solve
- Interest exists, e.g. Bluetooth remote control…
- People want to connect with people, not with data.
  - Email, IM, SMS, family photos, weblogs…

What should you expect?

Expect more diversity

- Goal: more customers, more reasons to buy your software

Expect managed diversity

- More platform APIs instead of device-specific ones
- Move to higher level software: not as constrained
- Less locked in to old-style PDA concepts

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Schedule for today

Palm OS in the Enterprise SPEAKER: ***Robert Hayes***
HANDSPRING: Developing and Optimizing your Applications for the Treo 600 Smartphone SPEAKER: ***Robert Gosselin***
BREAK
Developing Bluetooth and WiFi apps SPEAKER: ***Patrick Vuichard***
Increasing Compatibility and Performance in Palm OS 5 SPEAKER: ***Owen Emry***
Meet Our Partners Roundtable MODERATORS: ***PalmSource and Partners***

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Schedule tomorrow

The Big Picture: The Future of the Mobile Data Market SPEAKER: ***Michael Mace***
TAPWAVE: A New Mobile Entertainment Platform to Showcase Your Talents SPEAKER: ***Joyce Morrell and Ryan Robertson***
Mobile Operator and Infrastructure Provider Roundtable: Winning Wireless Applications SPEAKER: ***Albert Chu***
Becoming More Competitive with New Platform Technology SPEAKER: ***JB Parrett***
Publishing Palm Powered Software: What Works, What Doesn't. SPEAKER: ***Larry Berkin and Special Guests***
Palm OS 6 Sneak Peek SPEAKER: ***David Fedor***

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Q & A